

Rakumako Moloko

Final Year Student | AI Developer | AI Engineer | Data Scientist | Machine learning Engineer

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SUMMARY

I am a 4th-year BCom Data Science student at Belgium Campus with hands-on experience across software development, machine learning, and AI systems. I have practical exposure from roles at Softelsie Solutions, Nedbank, and Geeks4Learning, working on everything from predictive models to production AI pipelines. I am passionate about building intelligent systems that make businesses more efficient and cost-effective, with a particular interest in how AI is applied within financial institutions and enterprises. I thrive at the intersection of data, software, and applied AI.

EDUCATION

Bachelor's Degree in Bachelor of Computing (Data Science)

Belgium Campus ITVersity | 2023 – 2025

Currently Pursuing: Honours degree In Bachelor of

Computing (Data Science)

Belgium Campus ITVersity | 2026

National Senior Certificate (Matric) Mathew

Phosa High School | 2022

WORK EXPERIENCE

Softelsie Solutions PTY (LTD) Randburg

Software Tester (Learnership) | Jan 2025 – Mar 2025

- Performed manual testing of applications to identify defects and ensure software quality.
- Utilized ticketing systems to log, track, and manage reported issues, providing clear and detailed bug reports to development teams.
- Contributed to the development of a facial recognition application using C#, OpenCV, and related libraries.
- Gained practical experience in collaborative teamwork within an Agile development environment.

Nedbank Group – Technology Programme

IT Trainee - Digital Transformation Programme (Contract) | May 2025 – Aug 2025

- Completed comprehensive assessments in Machine Learning, TensorFlow, PyTorch, and advanced AI concepts

- Developed Anti-Money Laundering (AML) detection system for proposal using ML to identify suspicious transactions
- Participated in enterprise-scale digital transformation events
- Exposure to bank ethics and environmental adaptability

Geeks4Learning, Sandton Rivonia

AI Development Learnership | April 2026 – Present

- Contributed to a multi-agent AI pipeline project, helping automate workflows and reduce manual effort across the team
- Integrated LLM and RAG capabilities into the system, improving the quality and relevance of AI outputs for stakeholders
- Improved system reliability and scalability through database upgrades and asynchronous communication setup
- Implemented human-in-the-loop mechanisms to safeguard output quality before reaching production
- Delivered internal training content, supporting the company's learner development programme
- Completed training in Python, SQL, Computer Vision, MLOps, Machine Learning, and AI system design
- Participated in Agile sprint cycles, contributing consistently to team delivery and project milestones
- Acquired AI Development certifications along with technical training
- Attended all workshops including Soft skills training

TOP PROJECTS

1. Job Linked AI | React · FastAPI · PostgreSQL · Python · *Live Project*

Built a full-stack AI-powered job matching platform connecting South African job seekers with recruiters
 Implemented an AI candidate ranking and screening engine using ML scoring to automate and improve hiring decisions
 Integrated Paystack payment processing with subscription tier management for monetisation
 Deployed end-to-end on Railway, Vercel, and Cloudflare R2, delivering a production-ready SaaS product
 Features a recruiter analytics dashboard, automated CV screening, and in-platform messaging system

2. 3-Agent AI Pipeline | Python · LLMs · PostgreSQL · Kafka · Geeks4Learning

Built a multi-agent system with an orchestrator, code agent, and verifier agent to automate intelligent task execution.
 Integrated local LLM support via Ollama and a RAG pipeline to enhance response accuracy and relevance
 Migrated the database from SQLite to PostgreSQL and configured Kafka for reliable asynchronous agent communication
 Implemented human-in-the-loop oversight to ensure output quality before actions are committed
 Delivered a dashboard UI for real-time pipeline monitoring, improving operational visibility for the team
 Reduced manual intervention in repetitive workflows, enabling the company to scale AI-assisted processes more efficiently

3. RAG Banking Support Assistant | Python · ChromaDB · Ollama · LangChain · Geeks4Learning

Built a Retrieval-Augmented Generation (RAG) pipeline from scratch to answer banking policy questions using only grounded, cited context from a 10-document corpus
 Designed a document ingestion pipeline that chunks markdown documents by section heading, embeds each chunk using nomic-embed-text via Ollama, and stores vectors in a persistent ChromaDB database
 Engineered a system prompt with safety guardrails including PII protection, prompt-injection resistance, and explicit refusal behaviour for out-of-scope questions

Evaluated the pipeline against 12 test questions including deliberate trap questions, producing a scored results log
Stack: Python, ChromaDB, Ollama (llama3, nomic-embed-text), local vector search

ADDITIONAL COURSES AND ACHIEVEMENTS

ALX Africa - AI Career Essentials

Kaggle Certifications:

- Intermediate Machine Learning
- Deep Learning Specialization
- Python Programming
- Machine Learning Explainability (MLE)
- Feature Engineering
- Advanced SQL
- Game AI and Reinforcement Learning
- Data Visualization
- Time Series

2. Driver's License (Code 10)

SKILLS

Languages & Frameworks: Python, R, C#, SQL, React, FastAPI, TensorFlow, Keras, Scikit-learn, SQLAlchemy, LangChain, Ollama

Data & AI: Machine Learning, Computer Vision, LLM Integration, RAG Pipelines, Multi-Agent Systems, Anomaly Detection, Feature Engineering, Model Evaluation

Databases & Infrastructure: PostgreSQL, SQLite, Kafka, Cloudflare R2, Railway, Vercel

Tools & Platforms: Power BI, Figma, Git, Kaggle, Ollama

Practices: Agile/Scrum, Unit Testing, MLOps, Human-in-the-Loop Design, Software Development, Data Analytics

Soft Skills: Problem Solving, Critical Thinking, Adaptability, Attention to Detail, Team Collaboration, Communication, Self-directed Learning, Time Management

LEADERSHIP AND ACTIVITIES

- Grade 12 Math and Physical Science Tutor
- Snooker / Pool
- Chess
- Soccer

REFERENCES

Available upon request